

Inside the Atom: Protons, Neutrons and Electrons

Answer key

A. The particles

1. Charge, mass and location.

Proton: charge +1, mass about 1 u, in the nucleus.

Neutron: charge 0, mass about 1 u, in the nucleus.

Electron: charge -1, mass about 1/1836 of a proton, around the nucleus.

2. Atomic number and mass number.

Atomic number = number of **protons** (identifies the element). Mass number = number of **protons + neutrons**.

B. Counting particles

3. Protons, neutrons, electrons.

	Protons	Neutrons	Electrons
(a) F	9	10	9
(b) K ⁺	19	20	18
(c) S ²⁻	16	16	18
(d) Al ³⁺	13	14	10
(e) Zn	30	34	30

4. Write the symbols.

(a) 26 protons = iron; $A = 26 + 30 = 56$; 24 electrons gives 2+: ${}^{56}_{26}\text{Fe}^{2+}$

(b) 8 protons = oxygen; $A = 8 + 10 = 18$; 10 electrons gives 2-: ${}^{18}_8\text{O}^{2-}$

C. Isotopes

5. Neutrons in hydrogen isotopes.

Hydrogen-1: **0**; hydrogen-2: **1**; hydrogen-3: **2** (each has 1 proton).

6. Why isotopes behave the same chemically.

They have the same number of protons and therefore, as neutral atoms, the same number of **electrons**. Chemical behaviour depends on the electrons, not the neutrons.

D. Challenge

7. Why 35.45?

35.45 is the **weighted average** mass of chlorine's naturally occurring isotopes (mainly chlorine-35 and chlorine-37). Mass numbers are whole-number counts for individual isotopes.